

VBWT Shiny App – Feature Roadmap Summary

Goal:

Transform the app into a “Where should I go birding?” decision tool.

Core Features:

- Species search (name + 4-letter code)
- ZIP-based hotspot recommendations
- Ranking using detection rate (not raw sightings)
- Relative frequency (“X× more likely”)
- Recent activity layer

Timing & Seasonality:

- Weekly/monthly frequency charts
- Peak month insights
- Seasonal comparisons

Habitat Intelligence:

- Assign habitat using NLCD
- Filter hotspots by habitat
- Show habitat composition

Advanced Analytics:

- Bias reduction (complete checklists, smoothing)
- Confidence levels
- Hidden gems
- Stability metrics
- Specialty species index

Visualization:

- Interactive map (size = activity, color = likelihood)
- Ranked hotspot table
- Charts (seasonality, habitat)

Species Experience:

- Species photos (with attribution)
- Species profile pages

VBWT Integration:

- Site profiles
- Trail grouping
- “Best VBWT sites” recommendations

Outputs:

- PDF reports
- CSV downloads
- Trip summaries

UI Strategy:

- Desktop: sidebar + map + details
- Mobile: stacked layout

Data Strategy:

- EBD (quarterly updates)
- eBird API (recent data)
- NLCD (habitat)

Development Phases:

Phase 1: Search, ranking, ZIP, charts

Phase 2: Relative frequency, recent data, photos

Phase 3: Habitat, VBWT profiles, reports

Bottom Line:

A smart, location-based birding assistant for Virginia.